**VICTORIA JUNIOR COLLEGE**

**SUGGESTED SOLUTIONS TO 2023 PHYSICS H2 PRELIM PAPER 3**

Q1(a)(i)

Method 1: Using equations of motion and taking downwards to be positive:

$v^2 = u^2 + 2as = (1.5)^2 + 2(9.81)(0.65)$ [1]

Final speed = **3.9 m s$^{-1}$** [1]

**OR**

Method 2: Using conservation of energy

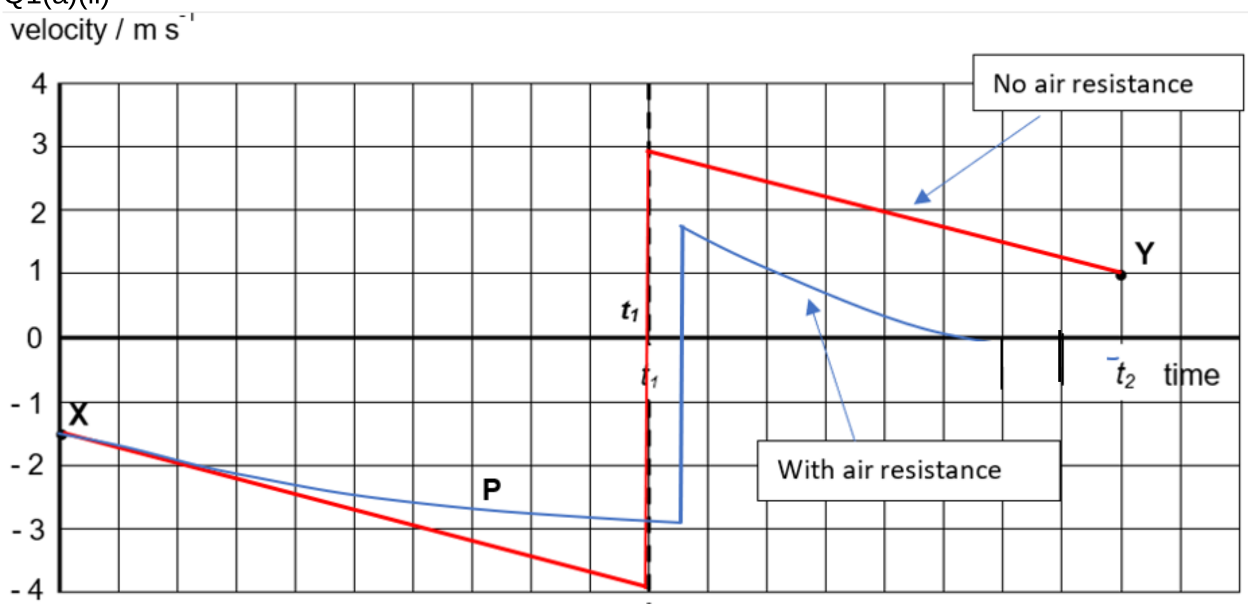
Loss in GPE = Gain in KE

$mgh = \frac{1}{2}m\left(v^2 - u^2\right)$

$9.81(0.65) = \frac{1}{2}\left(v^2 - 1.5^2\right)$ [1]

Final speed = **3.9 m s$^{-1}$** [1]

Q1(a)(ii)




- Straight line from X to – 3.9 m s$^{-1}$ [1]
- Gradient straight line from $t_1$ to $t_2$ must be parallel to line in first segment [1]
- Line at $t_1$ may join the two straight lines but its width must be narrow. [1]

Q1(a)(iii)
The speed of the ball after rebound is less than the speed just before impact. The ground/Earth is assumed to be stationary, hence there is a loss in the kinetic energy of the system during the bounce. [1]
The bounce is inelastic. [1]

Q1(b)
- Downward motion: gradient gets less steep with time and is a curve. [1]
- The ball takes a longer time to strike the ground [1]
- Upward motion: speed of rebound should be smaller and the velocity-time curve for upward motion of the ball should cover a smaller area representing a lower maximum height reached [1]